

Bluestock Mutual Fund Analytics Capstone

Final Project Report

1. Executive Summary

This project presents a complete mutual fund analytics pipeline covering ETL, data cleaning, SQLite database design, exploratory data analysis, performance analytics, advanced risk analytics, investor cohort analysis, recommender logic, and an interactive Power BI dashboard.

2. Project Objectives

- Clean and validate mutual fund NAV, AUM, SIP, transaction, performance, benchmark, and portfolio datasets.
- Design a SQLite star schema for structured analytics.
- Perform EDA using 15+ charts and document important business insights.
- Compute CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error, Maximum Drawdown, VaR, and CVaR.
- Build an interactive Power BI dashboard with four analytical pages.
- Develop advanced analytics including investor cohorts, SIP continuity analysis, sector concentration, and fund recommender.

3. Data Pipeline and Cleaning

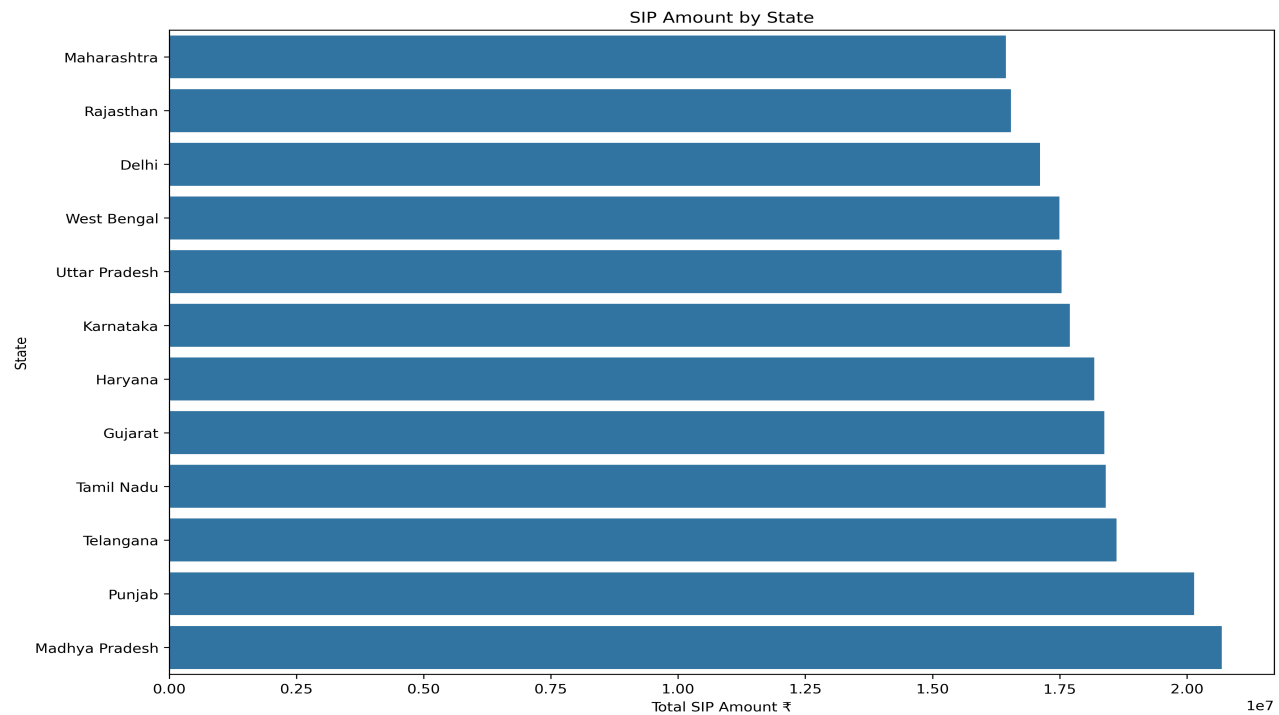
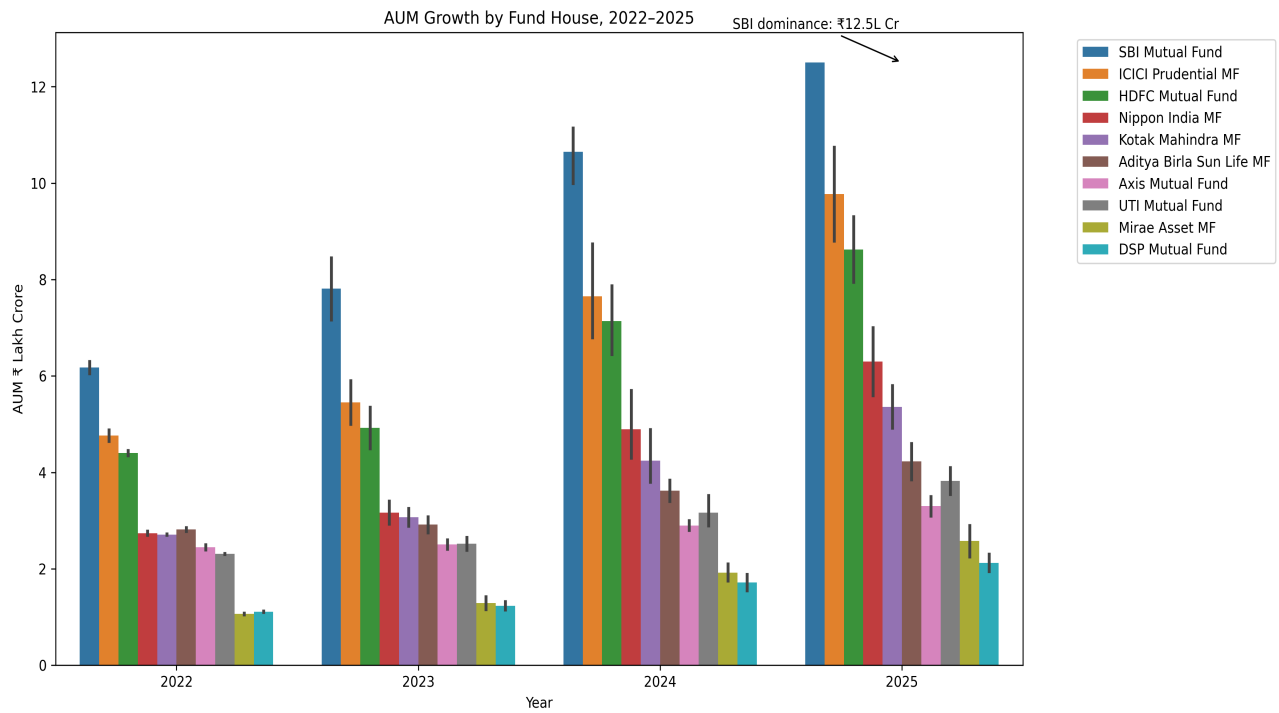
The ETL pipeline loads raw CSV datasets, validates schema consistency, parses date columns, removes duplicates, validates positive NAV and transaction amounts, standardizes transaction types, checks KYC status values, and creates cleaned CSVs in the processed data folder.

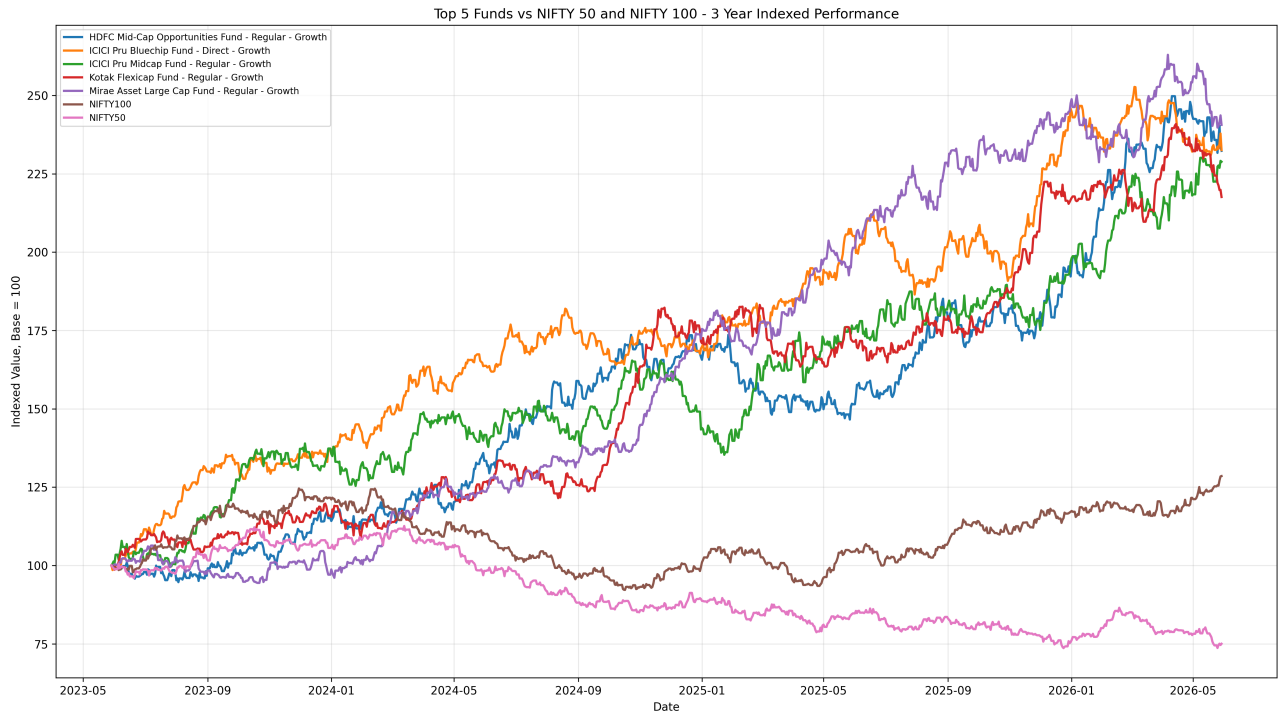
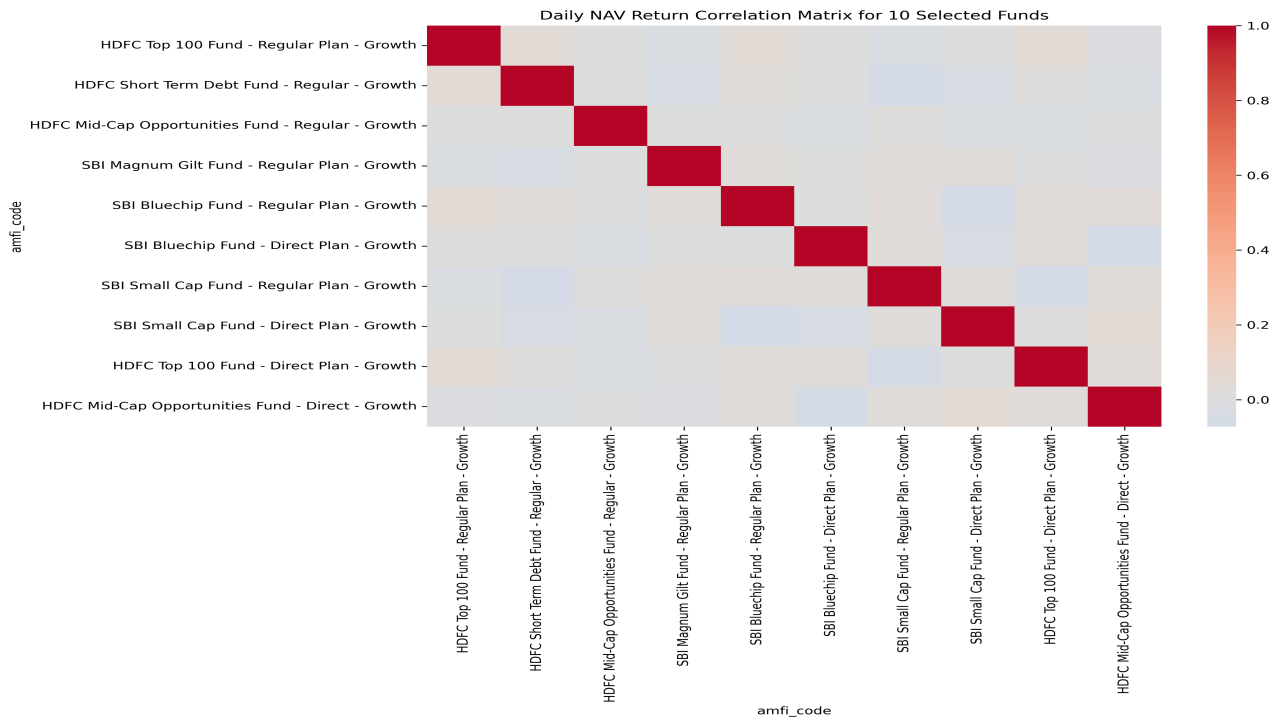
4. SQLite Database Design

A star schema was designed using dimension tables such as `dim_fund` and `dim_date`, and fact tables such as `fact_nav`, `fact_transactions`, `fact_performance`, and `fact_aum`. SQL schema and analytical queries are stored in the `sql` folder.

5. EDA Highlights

- NAV trends show broad growth across schemes with visible market movements during 2023 and 2024.
- SIP inflows show strong growth from 2022 to 2025.
- AUM analysis highlights dominance of major AMCs.
- Investor analytics shows transaction concentration across states, age groups, and city tiers.
- Sector allocation and category inflow charts reveal portfolio and market preference patterns.





6. Performance Analytics

Performance analytics included CAGR computation, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Tracking Error, and Maximum Drawdown. A composite fund scorecard was created using weighted ranks across return, risk-adjusted return, alpha, expense ratio, and drawdown.

Top 5 Funds by Composite Score

Scheme	Fund House	Category	Sharpe	Score
Mirae Asset Large Cap Fund - R	Mirae Asset MF	Equity	1.068	86.25
ICICI Pru Midcap Fund - Regula	ICICI Prudential MF	Equity	0.883	82.875
Kotak Flexicap Fund - Regular	Kotak Mahindra MF	Equity	0.966	82.0
HDFC Mid-Cap Opportunities Fun	HDFC Mutual Fund	Equity	0.808	80.75
ICICI Pru Bluechip Fund - Dire	ICICI Prudential MF	Equity	0.715	79.375

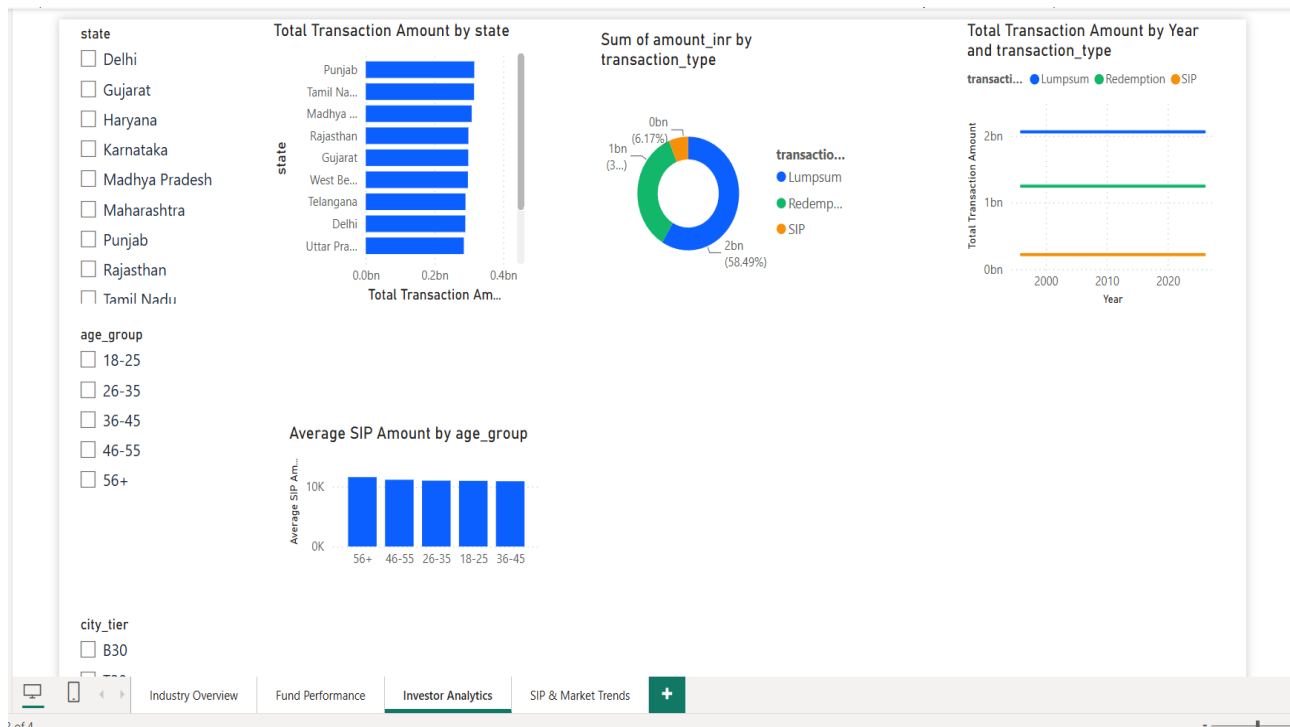
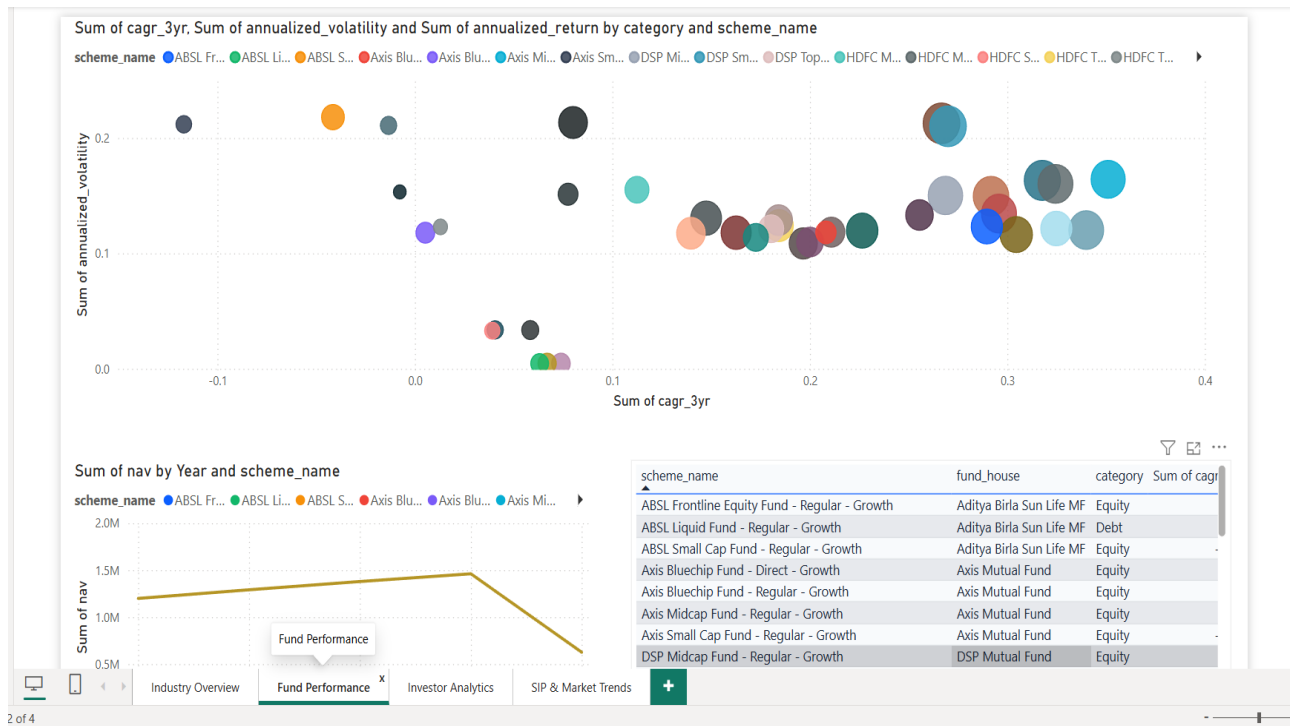
7. Advanced Analytics

- Historical VaR and CVaR were computed at 95 percent confidence for all schemes.
- Rolling 90-day Sharpe Ratio was plotted for key funds.
- Investor cohorts were grouped by first transaction year.
- SIP continuity analysis identified investors with gaps greater than 35 days.
- A simple recommender ranks funds by Sharpe Ratio within the selected risk appetite.
- Sector HHI concentration was calculated to identify concentrated equity portfolios.

8. Power BI Dashboard

The Power BI dashboard contains four pages: Industry Overview, Fund Performance, Investor Analytics, and SIP & Market Trends. It includes slicers, KPI cards, line charts, bar charts, donut charts, heatmap matrix, scorecard table, and benchmark comparison visuals.





- category
- ☐ ELSS
 - ☐ Flexi Cap
 - ☐ Gilt
 - ☐ Hybrid
 - ☐ Large & Mid Cap
 - ☐ Large Cap
 - ☐ Liquid
 - ☐ Mid Cap
 - ☐ Sectoral/Thematic

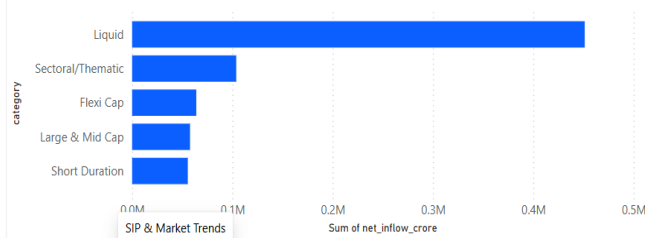
- index_name
- ☐ BSE_SMALLCAP
 - ☐ CRISIL_GILT
 - ☐ CRISIL_LIQUID
 - ☐ NIFTY_MIDCAP150
 - ☐ NIFTY100
 - ☐ NIFTY50
 - ☐ NIFTY500

Sum of sip_inflow_crore and Sum of close_value by Year



category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	346328	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Small Cap	34204	12392	46596
Total	706618	225621	932239

Sum of net_inflow_crore by category



9. Final Findings

- The mutual fund industry shows strong growth in SIP participation and folio count.
- Fund performance varies significantly by category, volatility, and risk-adjusted return.
- Composite scoring provides a more balanced ranking than return-only comparison.
- Investor behaviour differs across geography, city tiers, and age groups.
- Advanced metrics such as VaR, CVaR, drawdown, and HHI improve risk visibility.

10. Conclusion

This capstone demonstrates an end-to-end analytics workflow from raw mutual fund data to business-ready insights, risk analytics, fund ranking, recommender logic, and an interactive dashboard. The project is structured for reproducibility and professional submission.